

Prithwjit Ghosh

Data Scientist Senior Analyst | Accenture Technology

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🐙 GitHub | 📁 Portfolio | 🔗 LinkedIn

Data Scientist at **Accenture** with **2.5+** years of experience and a Master's from **IIT Kanpur**. Delivered analytics solutions to **six+** global clients across sales, finance and operations. Built production-grade models for accounts receivable forecasting and late payment prediction, cash flow, and sales forecasting. Skilled at translating insights into executive-facing Excel and Power BI dashboards and communicating effectively with leadership. Collaborative team player with a strong focus on delivering high-quality client solutions.

Professional Experience

July 2023 – Present

📊 Sales and Guest-Count Forecasting

- » Developed an ensemble forecasting model for *sales* and *guest-count*, projecting **48 months** ahead for business planning.
- » Integrated macroeconomic indicators such as *unemployment rate* and *price index* to enhance precision.
- » Implemented multiple models including **Prophet**, Theta, MSTL, LGBM, and Naive, totaling over **450** forecasts
- » Designed an *evaluation vector* incorporating **trend** and **seasonal** MAPE for a unique performance assessment.
- » Optimized horizon-based model selection through minimization of the **Euclidean norm** for reliability.
- » Classified the models into *overpredictive* and *underpredictive* categories to enhance the power of ensembling.
- » Incorporated pre- and post-processing techniques with different ensemble metrics to balance out the fluctuations.
- » Applied exponentially weighted *smoothing* over six months to stabilize forecast fluctuations for longer horizons.
- » Delivered accurate forecasts achieving **1–3%** MAPE for 24 months and **2–5%** for 48 months for both targets.

📊 Intelligent Collections 3.0

○ Accounts Receivable (AR) Forecasting

- » Focused on short-term account receivable forecasts, typically 1 month and up to 6 months for business planning.
- » Designed models across client segments including SMA, EWMA and business-specific forecast filters.
- » Implemented advanced time series models like ARIMA, SARIMA, **Theta** and **MSTL** for pattern recognition.
- » Developed ML and DL models such as XGBoost, LightGBM, CatBoost and *Temporal Convolutional Network*
- » Achieved forecast errors within **2–5%** across all AR buckets — *Not Yet Due*, *Current Due* and *Over Due*.
- » Built and deployed an interactive **Power BI** dashboard to present findings effectively to leadership and client demo.

○ Late Payment

- » Conducted deep customer profiling and entity analysis to predict future late payers with higher precision.
- » Trained **XGBoost** models using top *50 features* to predict late chance and duration at due date and due month levels.
- » Achieved AUC-ROC of **84%** for due-date late and **90%** for due-month late, with **5-day** MAPE for late days.
- » Developed a *risk score* and *risk categorization* combining all the predictions with a *quantile calibration*.
- » Delivered **38%** reduction in *overdue amounts*, **15%** lower **open AR** and **12%** higher *collections*.

📊 Cash-Flow Forecasting

- » Delivered end-to-end B2C cash flow forecasting with *daily*, *weekly* and *monthly* projections for 12 months.
- » Processed 90 GB (500 million rows) of raw data from **6 million** active customers using **Polars** for segmentation.
- » Built meta-learning models that improved Cash-In forecast accuracy to **98%** (↑ from 70%) using boosting approach.
- » Improved Cash-Out forecast accuracy to **93%** (↑ from 64%) through adaptive time series and ML model stacking.
- » Consolidated the complete analysis into an interactive fully functional **Power BI** dashboard for client presentations.

MSc Final Semester Project

May 2023 – July 2023

📊 Classification using Robust DD Classifier for Cellwise Contaminated Outliers | Mentor - Dr. Subhajit Dutta

- » Identified and estimated *cellwise outliers* from multivariate data using 2SGS and MCD estimator (rather than casewise).
- » Transformed data into a *k-dimensional space* for k-class classification problem while preserving class separation.
- » Evaluated multiple ML models, including *Logistic Regression*, *KNN*, *Random Forest* and *SVM* with various kernels.
- » Proposed a **Robust 2SGS classifier** outperforming benchmarks across simulated cases (*exp*, *non-exp* and *mixture family*)
- » Achieved moderate to decent improvements on real-life benchmark datasets that had cellwise outliers.

Skills

- **Programming Languages:** Python, SQL, R, $\LaTeX$, Bash-Scripting
- **Software:** MS Excel, MS PowerPoint, Minitab, Maple
- **Deployment Tools:** Power BI, Streamlit, AWS, Git
- **Libraries:** Numpy, Pandas, Polars, DuckDB, Matplotlib, Scikit-learn, XGBoost, Darts, Tensorflow, Keras
- **Domains:** Data Science, Data Analysis, Predictive Modelling, Statistics, Time Series, Inference, Business Intelligence

Education

🎓 M.Sc in Statistics || Indian Institute of Technology, Kanpur (IITK) || CGPA: 8.9

August 2021 - July 2023

- * A*/A : 7 courses (Multivariate analysis, Simulation, Bayesian Analysis)
- * *Final Project:* Robust Depth-Depth Classifier for Cell-wise Contaminated Outliers
- * Received grade A (**highest**) in the M.Sc. final project.

🎓 B.Sc in Statistics || Bidhannagar College || CGPA: 9.99

June 2018 - July 2021

- * A*/A: All courses (total 26)
- * Secured All India Rank (**AIR**) **7** in Joint Admission test for Masters (**JAM**) on Mathematical Statistics (MS).
- * Selected to participate in **MTUSS (2019)**, one of the most prestigious mathematics workshops, organized by *National Board of Higher Mathematics (NBHM)*.

Certifications

Data Analysis with R (Google) *Machine Learning Specialization (Stanford University)* *Python 101 for Data Science (IBM)*